BENCHMATE · TEAM 103 · MAKER DAY 3

Five-Minute Script

12 slides, about 700 spoken words, paced at 140 words a minute. The clock column is when each slide should come up.

Dotted = cut first if you are running long (saves about 8 seconds in total).

CLOCK	SLIDE	SAY
0:00	**1** Cover	Hi, we're Team 103. This is BenchMate: scalable, safe training, from expert to novice.
0:05	**2** Customer & Wedge	BenchMate turns an expert's recording of a hands-on task into a step-by-step tutorial. Our customer is any organization that trains people for hands-on work. The order is prove it, earn trust, then sell. Our wedge is novice makers at MakerLAB, where we prove it works. Next come technical colleges, an easier place to earn trust and early data. Then manufacturers' employee training teams, the paying customer, who we still need to interview.
0:30	**3** Tallest Tales	We listed 29 hypotheses and scored each for doubt and importance, one to three. Multiplied, that is the risk factor. Four score the maximum of nine: that learners finish tasks 30 percent faster, that organizations will share proprietary SOPs, that the AI answers fast enough for real-time help, and that learners trust its guidance. Each is something we have little evidence for and cannot do without. The fifth, at six, is that AI can turn raw video into instructions.
1:03	**4** Three Experiments	So we designed three experiments, each with a threshold set before the run. One, video to instructions, is a technical benchmark: four uncut videos against expert labels, passing at 80 percent accuracy with no critical omissions. Two, learner trust, is a Wizard-of-Oz prototype: eight novices work through a simulated assembly, and we need three in four to follow the AI. Three, time saved, is an A/B pilot: BenchMate against conventional training, timed to correct completion. We need 30 percent saved.

CLOCK	SLIDE	SAY
1:42	**5** Path	Our scarce resource is team hours, so we order by risk reduced per hour, within constraints. Extraction and the safety check go first, 19 hours, because every later test needs their output. They finish November 17. The latency benchmark, four hours, waits for a stable API and finishes on the 18th. Then the trust test, also the 18th, and the MakerLAB pilot, finishing November 19, which needs lab approval and filming consent. If lab access slips, the first three steps still run.
2:12	**6** Experiment 1	Experiment one is underway, on two videos so far. What holds: every fact we checked for was found, and both expected gaps were flagged. What breaks: on the uncut video the system produced 22 steps where the expert reference has seven, so its step timing scores zero against a target of 0.8. The content is right, but the steps are split too finely. These are keyword checks so far, not expert grading.
2:40	**7** Experiment 2	Experiment two starts with safety, because the trust test needs guidance an expert has checked. Across two runs, the tutorial kept 100 percent of the expert's original warnings and added four to six more. Where the video and the SOP disagreed, it flagged six conflicts to the expert instead of guessing. One added caution was wrong, so expert review stays in the loop.
3:08	**8** Experiment 3: Expert Time	Experiment three: a first run is complete, and it needs more participants. On the expert side, one soldering recording took four and a half minutes to film, then about two minutes of work per step in BenchMate: 28 minutes for 14 steps. Editing it by hand would take 40 minutes to an hour.
3:28	**9** Experiment 3: Learner Time	On the learner side: two first-timers, the same soldering task. With a YouTube tutorial it took 19 minutes. With BenchMate, five minutes fourteen. That is 72 percent less time, against our 30 percent threshold. It is one learner per condition, and YouTube is only a first benchmark, so this is a direction, not yet a pass.

CLOCK	SLIDE	SAY
3:51	**10** Experiment 3: Cost & Quality	Now scale to 20 learners. An expert teaching each one in person spends about an hour and a half. With BenchMate the expert spends about 33 minutes, once. That saves 58 minutes, about 64 percent, and we saw no discernible change in soldering quality.
4:12	**11** What Changed	So what changed? On extraction, facts hold but steps over-split, so fixing how steps are split is what we build next. On trust, safety guidance survives editing, but one wrong caution means expert review stays in the product. On value, the savings look large, so we run the MakerLAB pilot, then test against the training buyers actually use. And on the canvas, technical colleges now follow MakerLAB, and our buyer is manufacturers' training teams. These are small samples: directional evidence, not yet a pass.
4:55	**12** Thank You	Thank you. We'd be glad to take your questions.

DELIVERY

- **Handovers.** The natural breaks are after slide 5 (plan → results) and after slide 7 (Experiments 1–2 → Experiment 3). Three speakers fit these blocks.
- **Checkpoints.** You should be starting Experiment 1 at 2:12 and Experiment 3 at 3:08. If you reach slide 8 after 3:20, drop every dotted sentence that is left.
- **Point, then speak.** Each results slide has one large number. Say that number first and let the slide carry the rest.
- **Keep the caveats.** "Keyword checks", "one learner per condition" and "one caution was wrong" take seconds to say and answer the first hard questions before they are asked.